William D. Boulton

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OBJECTIVE

I am a student at Purdue University looking for career and research opportunities in software development and cybersecurity to apply my interest in mathematics and computer science.

EDUCATION

Purdue University Expected Graduation: Spring 2027
Bachelor of Science – Computer Science Honors GPA: 3.95 West Lafayette, IN

- **Dean's List**: 2024–2026
- **Semester Honors**: 2024–2026

EXPERIENCE

Trail of Bits 2026
Intern - Research and Engineering Brooklyn, NY

- Developing an agent benchmarking and debugging suite
- Accelerated benchmark task development allowing the creation of targeted, high-quality datasets in minutes
- Implemented various debugging and optimization features to accelerate development of specialized agents (measured up to **30%** improvement from automated optimizers)

Trustworthy Software Ecosystems 2025 – 2026
Undergraduate Researcher Purdue University

- Co-authored **Debtrace**, a paper which traced the Debian software ecosystem to find discrepancies in package provenance
- Determined the Debian reproducibility project over-reports packages as reproducible by **27%**
- **Direct accepted** into ICSME 2026 (26.5% acceptance rate)
- Winner of ICSME IEEE TCSE **Distinguished Paper Award**

Purdue College of Engineering 2025
Summer Undergraduate Research Fellow (SURF) Purdue University

- Developing a prototype of a Trusted Builder Service to ensure security during the software rebuilding process
- Used a trusted application in **Arm TrustZone** to oversee and endorse software builds
- Developed components for SBOMctl to create trustworthy software supply chain infrastructure capable of preventing widespread supply chain compromises (i.e. SOLARBURST, JetBrains TeamCity compromise, NotPetya)

Intelligent Generation of Tools for Security (INGOTS) 2025
Undergraduate Research Assistant Purdue University

- Vulnerability research on **Android userspace applications and kernels**, primarily focused on recreating and debugging Chromium and Android kernel exploits
- Developed Android exploitation tools as part of the **DARPA INGOTS** research project

Siemens Digital Industries Software 2024
Career Start Intern Multinational (Japan, India, United States)

- Created and implemented a Mendix-powered web application that automated the interview scheduling process for the head of the Career Start Internship program
- Used **AWS** to analyze user statistics and develop solutions to increase efficiency for company websites

ACTIVITIES & PROJECTS

b01lers – Purdue Capture the Flag (CTF) Offensive Security Team

- Compete in weekly CTFs and attend/host weekly training sessions; developed challenges for **b01lersCTF** in 2025 and 2026
- Placed **top 10 in the United States** in 2024 and 2025 and **2nd place in MITRE eCTF** 2025
- Maintained b01lers server infrastructure used to host b01lersCTF and internal competitions and tools